



Design and Development of a Recycled Digital Ohmmeter Utilizing an AT Mega Microcontroller

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Abstract

A major problem that is being faced in Egypt and has been worsening over time is E-waste, because it doesn't just pollute a specific place in the air or water but, toxic materials are being released that leach into soil and underground water risking the health of both humans and wildlife. The chosen solution after a lot of research is proper recycling, which will be done by building the entire project from scratch from E-waste, furthermore that it's not thrown away or disposed wrongly, not only reducing E-waste but also encouraging creativity for Eco-friendly devices. This prototype was designed, constructed, and tested to meet our design requirements such as that the error percentage can't exceed 5%, the prototype to be fully recycled to be eco-friendly, and to have one electric part at least. The chosen prototype was a digital ohmmeter, the design requirements were efficiently satisfied by the prototype results as the average error percentage that was measured after calculating the resistance of 3 different resistors in 3 different trials was 3.2% which is much less than the 5% design requirement, it was found that the prototype sometimes gets results that were slightly better than actual multimeters that were used in school laboratories. Results and conclusion from the prototype ensured that the prototype was working well while the range of the resistance is limited correctly on the circuit itself.

Introduction

Several problems are faced by Egypt known as "The grand challenges" of Egypt, there are three main grand challenges that are connected to our solution which are "Reduce and adapt to the effect of climate change", "Address and reduce pollution", and "Recycle garbage waste for economic and environmental purposes", These challenges are all affected by the problem of "E-waste".

E-waste is made from electronic devices, mostly from throwing away old devices because it doesn't work anymore or from people wanting to get the newer version of this device which is done a lot in our time right now, More and more E-waste is being thrown each year as illustrated in (Graph 1).



Graph 1: E-waste generated over years

Hazardous materials are often found in E-waste devices, which are toxic materials that are released from the devices leaching into soil and underground water also polluting air and water when burnt or disposed of in the wrong way.

Many efforts were put on that topic to find a proper solution that can end this problem or at least reduce it to the minimum amount, disposing correctly was suggested which is a good because it doesn't pollute the environment, but it has various drawbacks as it's too expensive, very complicated, and needs a lot of machines and efforts which will affect the economic side of the country badly, another solution was suggested is to start using materials that are more reliable and don't need to be changed every year which is great since people won't throw away their phones but this will be too expensive and not everyone will be able to do that.

The selected solution must have specific design requirements to ensure efficiency and reliability. The design requirements that were chosen for our solution are for it to be eco-friendly, The error percentage must not exceed 5% to ensure accuracy and efficiency, moreover, it must be user-friendly so that anyone would be able to use it without the need of going through many instructions.

The solution chosen was to recycle E-waste materials from old devices to build new devices, recycling was found better than most of the other prior solutions because it reduces pollution in an easy and cheap way that doesn't require complex processes, saves valuable materials because most electronic devices contain materials such as gold, silver, and copper which are rare materials, furthermore the environment is harmed from the extraction of these materials from earth, it boosts economy since it supports the idea of circular economy where materials are being used again and again instead of being thrown, and the most important reason that we chose that solution for, is because it supports innovation and creativity since recycling these materials can be used to teach designing and engineering skills.

The chosen prototype was a digital ohmmeter, that was preferred over the analog ohmmeter because digital ohmmeters have more accurate and precise measures and it is also easier to read so the user-friendly requirement is checked, furthermore for the eco-friendly requirement to be checked, the entire project will be built from e-waste materials.

Materials & Methods

Table 1: Materials used in construction

Material	Quantity	Usage	Source	Cost	Picture
ATmega328PU (DIP-28)	1	Main Processing Unit	Power Supply	Recycled	
Resistors	7	Powering LED, Communication Protection, and Voltage Divider reference	Power Supply	Recycled	
I2C LCD Display (16x2)	1	Shows resistance values	Old Calculator	Recycled	

Capacitors	6	Filters ADC input noise for stable readings.	Computer Motherboard	Recycled	
16MHz Crystal	1	Clock Source	Computer Motherboard	Recycled	
LM7805 Voltage Regulator	1	5V Power Regulation	Computer PSU	Recycled	

Total Cost=Zero (All materials were recycled)

Methods:

The prototype went through 4 main construction stages, ranging from assembling the microcontroller to making the display and interface, as everything needed to be made with precision and accuracy.

The main stages of constructing the prototype were:

- 1) Assembling the Microcontroller where an ATmega328PU was salvaged from an old power supply. 16MHz crystal and 22pf capacitor was added for timing. Reset circuit was built from 10kΩ resistor and 10uF capacitor. FTDI programmer was used to apply the code via RX/TX pins.
- 2) The Power System was made using LM7805 voltage regulator which converts the current to 5V and the filtering capacitors put were 100nF and 10uF ensuring clean power.
- 3) The part of the Ohmmeter's Circuit was applying Voltage Divider where 5V were moved through the 10kΩ reference resistor then to the test points and GND. Then Arduino A0 pin was used to measure the difference between the reference and unknown resistor. The main calculation was put using Ohm's Law where $R_{Unknown} = 10k\Omega \times V_{out} / (5V - V_{out})$.
- 4) The last part was the Display, made out of L2C LCD which shows the resistance with Ω symbol. The unknown resistance needed to be measured was connected between test point A and the GND, connecting it later to A0/Some.

Safety Precautions were followed during the construction of the prototype to prevent and avoid any possible injury. Some of these safety precautions were wearing coats and gloves, working in the fabrication lab, staying always under supervision, and using resistors and voltage regulators to prevent excess of current flow; as it could result in emitting heat and causing harm to both, the prototype and the team.

Test plan:

To ensure the prototype's efficiency, whether the functional requirements and the delivers reliable measurements were achieved or not, it was tested in 4 trials using resistors with known values of 2kΩ, 5.1kΩ, 10kΩ, and 100kΩ, respectively. The device operated from a stable 5.00V supply since the resistance calculation depends directly on the accuracy of the voltage divider formed with the selected reference resistor. The result of the prototype in the measurement was compared to that of a digital multimeter to determine whether its performance justifies building the device instead of relying on a commercial multimeter or not.

Results

Negative results:

The first results that were measured had a reference resistor 10kΩ which was ideal for measures near that value but when the 100kΩ was put, this was way higher than the reference resistor which made the results 94.39kΩ, making the error 5.61% which exceeded the error's limit that was chosen. Moreover, although the effect is low, the resistance measurements were affected by the internal resistance of the reference resistor itself increasing the error percentage in an unpredictable way that can't be calculated.

Positive results:

After the reference resistor was changed each time to maintain an accurate and precise result without a big error and going through the test plan that was put, the results were very satisfactory. The results were taken from 3 different trials with 3 different resistors as shown in (Table 2).

First trial:

The resistor had a value of 2kΩ, the multimeter measurement was 1.952kΩ and the prototype in this trial had a result of 1.882kΩ to have an error of 5.9% for the prototype and a 2.45% for the multimeter.

Second trial:

The resistor had a value of 5.1kΩ, the multimeter measurement was 6.1kΩ and the prototype result was 5.00kΩ making the error of the prototype equal 1.96% and that of the multimeter equal 19.6%.

Third Trial:

The resistor that was used in the last trial had a value of 10kΩ, the multimeter measurement was 9.83KΩ and the prototype value was 9.826kΩ to have an error that is nearly 1.74% for the prototype and a 1.7% for the multimeter.

The average error that was calculated after all of these calculations was equal to 3.2% for our prototype which checked the required design requirement, and 7.91% for the multimeter which ensured the efficiency of our prototype and its reliability over laboratory devices under certain conditions.

Table 2: Results with reference resistor 10KΩ

Resistor Value	E-Waste Ohmmeter	Multimeter	Ohmmeter error	Multimeter error	Observation
2 kΩ	1.882 kΩ	1.951 kΩ	5.9%	2.45%	Multimeter closer
5.1 kΩ	5.00 kΩ	6.10 kΩ	1.96%	19.6%	E-Waste Ohmmeter closer
10 kΩ	9.826 kΩ	9.83 kΩ	1.74%	1.7%	Both nearly identical

Analysis

Many interconnected problems were faced by Egypt termed as "grand challenges". These challenges impact the country's economy and hinder its progress toward development, whether economically, socially, politically, or environmentally. E-waste is one of those challenges that has numerous side effects, whether direct or indirect. The amount of non-recycled e-waste, especially solid one, is very high in Egypt as shown in (Graph 2). About 90,000 tons of e-waste are generated annually, with only about 10-15% being formally recycled, risking harm to both animals and humans as a result of the contamination of soil due to the hazardous materials within the e-waste. Soil absorbs most of the heavy metals in the e-waste, such as lead, mercury, and cadmium, affecting soil health and degrading agriculture. The biggest reason for this is the lack of formal management, as most of the e-waste ends up in uncontrolled landfills or is processed using crude methods. This problem can be addressed by focusing on recycled projects. Those projects will make use of this disposed e-waste to make something beneficial, eco-friendly, and relatively cheap.

The E-Waste Recycled Ohmmeter addresses this challenge as it provides a low-cost, accessible solution that would especially help students, either in school or in university, in the labs by providing a high-efficiency ohmmeter in them.

This project showed significant results, as shown in the previous section, in comparison to other non-recycled commercial products. The voltage driver circuit is used by this project, which utilizes both Ohm's Law and Kirchhoff's Voltage Law as studied in (PH.2.03) and (PH.2.04), respectively.

Since Ohm's Law states that $V = I \cdot R$ and KVL The sum of all voltages around any closed loop in a circuit equals zero, The Voltage Divider Law would be derived using the inverse function studied in (MA.2.01) as $V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$. For this to be applied in the circuit, since the voltage of the microcontroller is 5V already and the reference resistor equals 10 kΩ, the final shape of the law would be

$$V_{out} = 5V \times \frac{R_{Unknown}}{10k\Omega + R_{Unknown}}$$

Now solving for R gives us $R_{Unknown} = 10k\Omega \times \frac{V_{out}}{5V - V_{out}}$. The power supply circuit of the prototype consisted of a voltage regulator and 2 decoupling capacitors. Those capacitors of (100nF, 10μF). The voltage regulator was used to dissipate the excess energy as heat demonstrating (ME.2.03) work principles where electrical work is converted to thermal energy. The work is calculated using the formula $W = F \cdot D \cdot \cos\theta$. The main principles of this device were **Joule's Law of Heating** which states that $P_{dissipated} = (V_{in} - V_{out}) \times I$, and the capacitors, as studied in (PH.2.05), filtered noise exploiting the equation $\tau = R \times C$, and its energy storage using the equation $E = \frac{1}{2} CV^2$ where E is the energy and C the capacitance. The prototype's results, shown in the previous section, were analyzed both the positive and negative.

In the **First Trial**, the measurement of actual value of the resistor was 2kΩ. However, when measured with both the multimeter and e-waste ohmmeter, it showed error percentages of 2.45% and 5.90% respectively. The error percentage can be calculated utilizing (MA.2.01) which was beneficial for the second time, using the formula $\frac{|Measured - Actual|}{Actual} \times 100\%$.

This shows that the multimeter had a closer reading to the actual value than that of the prototype. This can be a result of how close the reference resistor is to the actual value, as in the multimeter, the reference resistor was 2kΩ while that in our prototype was 10kΩ. Since the multimeter's reference was closer, it was expected that it'll have more accurate reading than ours.

In the **Second Trial**, the resistor, unknown to the prototype, had a resistance of 5.1kΩ, much closer to the value of the reference resistor than the previous trial. However, it was far away than the reference of the multimeter—of 20kΩ. This can be the main reason of the huge error between the multimeter's measurement and the actual value as there was a 1kΩ difference which is relatively high considering the actual value. However, the measurement from our prototype to this resistor was 5.0kΩ very close to the original value, showing accuracy of about 98%, perfect in comparison to the design requirement requiring the prototype to show error of 5% at most.

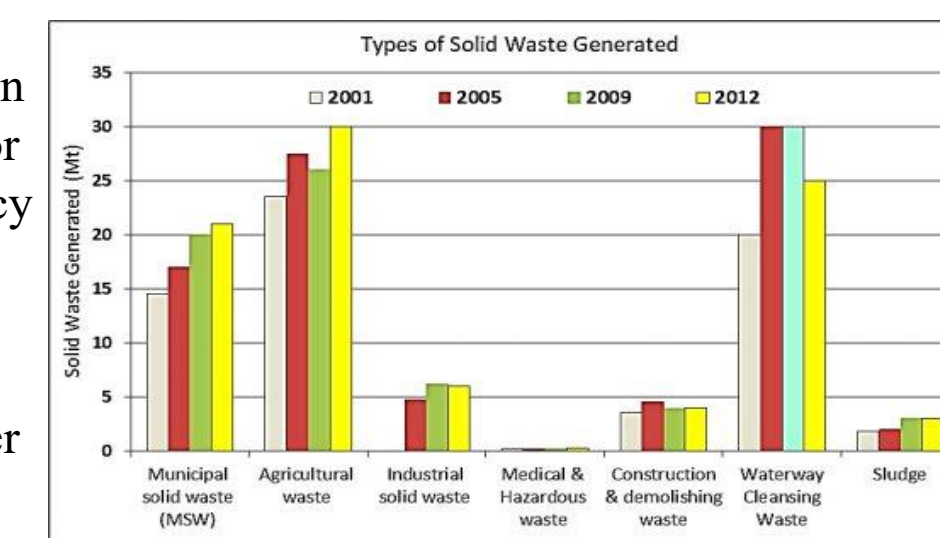
During the **Third Trial**, when measuring a resistor of value 10kΩ, both devices had nearly identical measurements as they are both close to the reference resistor. This trial had the most accuracy among them all for both devices as it was 98.26% for the ohmmeter and 98.30% for the multimeter. When analyzing this trial correctly, there is a slight advantage towards the multimeter. Although it had a further reference resistor, it maintained high accuracy compared to its previous trial.

In the Fourth Trial, the results were a bit odd.

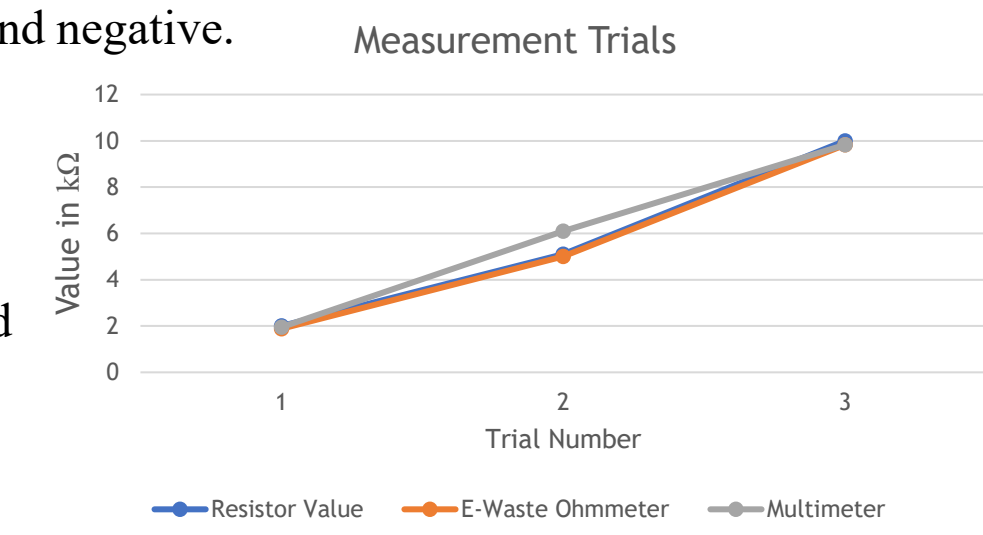
The actual value of the resistor, as shown in (Figure 3), was 100kΩ. The reference resistor of the Multimeter which is manually edited each trial, was 200kΩ and that of our prototype was 10kΩ. In this trial both devices showed high errors in comparison to the actual value as of the huge difference and the ohmmeter had a high error in it. It had an error percentage of 5.94% which is the most it can be, as the design requirements specify an error of ≤5%.

All of the data collected from this test and every test the prototype went through in general before this one was collected using (CS.2.01) and (CS.2.02). it helped in showing comparative measurements, calculating error percentages, and identifying the pattern that improves the accuracy when unknown resistor ≈ reference resistor.

To sum things up, the E-Waste Ohmmeter was considered successful. All the design requirements were achieved. Moreover, it integrated 7 learning outcomes helping us understand the curriculum itself beside extracurricular research concepts. This makes theoretical classroom concepts seen and utilized in real life, helping us not just in memorizing it, but understanding and applying the whole concept. Lastly, it would be a great way of facing the e-waste problem in Egypt.

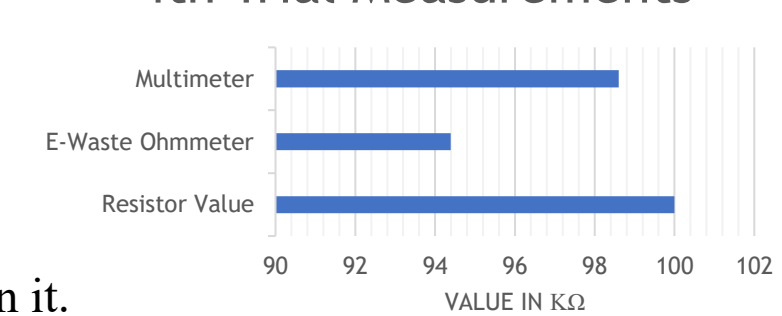


Graph 2: Types of solid waste generated



Graph 3: Measurement trials

4th Trial Measurements



Graph 4: 4th trial measurement

Conclusion

The increase in electronic waste and its environmental risks made us focus on making a project that solves this problem. The project is responsible for representing a practical method to recycle E-waste into new functional electronic devices. The prototype uses entirely reused electronic components from discarded devices, which is a locally rich waste material. We concluded that:

- The prototype demonstrates that salvaged ATmega microcontrollers can perform precise resistance measurements, achieving a 96.8% accuracy.
- The prototype successfully measured resistor values with a high degree of accuracy. In a key test with a 10 kΩ resistor, it measured 9.826kΩ against a multimeter's 9.83kΩ, demonstrating precise functionality.
- The prototype can be used as a fully functional digital ohmmeter achieving 96.8% average measurement accuracy (3.2% error) across a 2kΩ to 100kΩ range
- The entire assembly and validation process for the recycled ohmmeter was completed successfully, meeting the core design requirement of creating a functional, user-friendly device from E-waste.
- Although the 100 kΩ measurement showed a 5.61% error, this result revealed the system's optimal operating range and provides clear direction for future improvements, such as auto-ranging reference resistors.
- The prototype operated safely throughout all tests, with no failures or increased error rates even when measuring resistances across a wide range, confirming its robustness for educational and diagnostic applications.

Recommendations

It's a fact that nothing in the whole world is perfect. However, these are some recommendations that can improve the project efficiency in most aspects, which are:

- Adding averaging algorithms which help approximate the values of multiple testing cases.
- Adding calibration system into the Ohmmeter to ensure its accuracy by measuring the resistance of the wires and calculating the outputs right.
- Implementing auto-ranging by switching between multiple reference resistors (1 kΩ, 10 kΩ, 100 kΩ) to maintain high accuracy across a wider resistance span.
- Using diode clamping circuit to fix the output to a specific limit.
- Using trimmer potentiometer which works on "set and forget" mechanism that is used in zeroing the ohmmeter and reducing error
- Using kelvin test probes "specialized test leads" to detect and measure small changes in resistance
- Incorporating a battery-level indicator and low-power sleep mode to extend operating time when used in portable, field-testing situations.

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